A 130-Snapshot Audit of the Physical History Realization

Stable Polar Packets, One-Sided Accuracy, and a Two-Sided Cocycle Failure

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Abstract

RH-172 and RH-173 provide exact finite formulas for the reset-memory history factor $M_t = F_t^* F_t$, its canonical polar packet, and the rectangular update $F_{t+1} = T_t F_t$. We now test those formulas on every reset snapshot in the five-scale RH-151 atlas: 130 packets and 120 consecutive updates across the left and right channels.

The structural identities are numerically stable. The maximum Gram residual is 8.30×10^{-16}, the maximum cocycle residual is 9.36×10^{-16}, and an SVD polar implementation keeps all 130 isometry defects below 3.57×10^{-15}. Direct evaluation of $F_t U_t \Lambda_t^{-1/2}$ is less reliable because the selected Gram condition number reaches 3.61×10^{12}: only 103 of 130 direct formulas have isometry defect at most 10^{-8}, with maximum defect 1.57×10^{-5}.

Packet coherence is qualitatively different. Ninety-five of 120 primal relative residuals are at most 0.10, but every adjoint relative residual is larger than 0.25; the observed range is $[0.3233, 0.99998]$. Consequently zero updates satisfy the symmetric 0.25 two-sided gate. The transported reset subspace has median distance 0.1873 from the new reset packet, maximum distance 0.99999965, and 23 distances at least 0.90.

Thus the canonical finite history realization is real and stable, but the most direct consecutive-reset reducing-bundle hypothesis is not supported by these frozen data. This is a floating finite-data negative result, not an all-level no-go theorem. It leaves open a different packet schedule, a lagged or block cocycle, a cyclic closure, and every direct transfer-space construction. No physical Riesz interface or downstream Hilbert–Polya claim is closed.

1 Question and data provenance

RH-151 certified 130 independently reset source-memory packets using Arb matrix balls and a clock-rank spectral gap [1]. RH-172 then proved that every such packet has a canonical finite history realization [2], while RH-173 derived the exact rectangular history cocycle and warned that Gram optimality alone does not imply invariance [3]. The present paper asks the concrete physical question: do the archived reset packets nevertheless behave as an approximately reducing cocycle bundle on the five frozen models?

The models are rebuilt by the same RH-77/RH-94 routines used upstream [4, 5]. The left channel acts on the fine bulk and the right channel on the coarse transposed bulk. The source-memory space is the column space of the corresponding source matrix. The forgetting parameter is

$$\eta = \frac{1}{512}.$$

No eigensolver output from the RH-151 JSON is silently promoted to exact data; the states, memories, and packets are recomputed in double precision.

σ	left dim.	right dim.	source cols.	rank	endpoint	snapshots
0.16	32	16	16	4	4	10
0.08	64	32	32	5	6	14
0.04	128	64	64	6	11	24
0.02	256	128	128	6	17	36
0.01	512	256	256	7	22	46
total						130

Each scale contributes two channels. Removing the ten initial snapshots leaves 120 consecutive updates.

2 Audited objects

For each channel, write $S_t = A^t S_0$, normalize $\widehat{S}_t = S_t / \|S_t\|_F$, and set

$$F_t x = (\widehat{S}_t x, \sqrt{\eta}\widehat{S}_{t-1} x, \ldots, \eta^{t/2}\widehat{S}_0 x).$$

The recomputed memory is $M_t = F_t^* F_t$. If U_t contains the top clock-rank eigenvectors and $U_t^* M_t U_t = \Lambda_t$, two floating realizations are compared:

$$V_t^{\mathrm{dir}} = F_t U_t \Lambda_t^{-1/2}, \tag{1}$$

$$F_t U_t = L_t \Sigma_t R_t^*, \qquad V_t = L_t R_t^*. \tag{2}$$

Equation (1) is the exact spectral formula. Equation (2) evaluates the same polar factor by a thin SVD and is used for all subspace and cocycle diagnostics.

For an update $t-1 \to t$, let

$$r_{t-1} = \frac{\|S_{t-1}\|_F}{\|S_t\|_F}$$

and let T_{t-1} be the exact history cocycle. The transported old packet is $W_t = T_{t-1} V_{t-1}$. Its polar range is compared with V_t using the operator distance between orthogonal projections. We also compute

$$\epsilon_t^{\mathrm{pr}} = \frac{\|(I - V_t V_t^*) W_t\|}{\|W_t\|}, \tag{3}$$

$$\epsilon_t^{\mathrm{ad}} = \frac{\left\|(I - V_{t-1} V_{t-1}^*) T_{t-1}^* V_t\right\|}{\|T_{t-1}^* V_t\|}. \tag{4}$$

The denominators make the two quantities dimensionless. They are not outward-rounded upper bounds; they are diagnostics of the rebuilt floating matrices.

3 Exact-identity layer

The first result is an implementation validation of the exact theorems.

metric	minimum	median	maximum
$\|F_t^* F_t - M_t\|$	0	2.28×10^{-16}	8.30×10^{-16}
relative $\|F_t - T_{t-1}F_{t-1}\|_F$	2.20×10^{-16}	6.04×10^{-16}	9.36×10^{-16}
stable $\|V_t^* V_t - I\|$	4.86×10^{-16}	1.61×10^{-15}	3.57×10^{-15}
stable polar relative factorization	1.32×10^{-16}	1.77×10^{-15}	1.07×10^{-14}

All 130 stable polar packets have isometry defect below 10^{-12}. The history construction itself is therefore not the numerical weak point.

4 Conditioning and the direct inverse-square-root formula

The selected packet condition number

$$\kappa_t = \frac{\lambda_{t,1}}{\lambda_{t,r}}$$

ranges from 1.095 to $3.6048052854825 \times 10^{12}$, with median 1.721725×10^{6}. Such values are compatible with a certified packet gap: separation from the next eigenvalue does not imply that the smallest selected eigenvalue is large in absolute terms.

The direct formula (1) reflects this conditioning:

metric	minimum	median	maximum
direct isometry defect	1.27×10^{-15}	8.04×10^{-12}	1.57×10^{-5}
direct-to-stable frame residual	1.08×10^{-15}	4.03×10^{-12}	7.84×10^{-6}

Only 103 of 130 direct evaluations have isometry defect at most 10^{-8}. This is not a failure of the exact theorem. It is a reproducible warning that the physical implementation should use a polar SVD or a validated inverse square root, not unguarded columnwise division by $\sqrt{\lambda_j}$.

5 Packet transport: one-sided success is insufficient

The global update statistics are

metric	minimum	median	maximum
transported reset distance	1.85×10^{-5}	0.187289	0.99999965
primal relative residual	3.11×10^{-6}	0.0288333	0.721922
adjoint relative residual	0.323269	0.713027	0.999976
transported packet condition number	1.8373	16.2016	60.7507

The transported packet itself is not severely conditioned. The asymmetry is geometric: the old packet often maps close to the new packet, while the new packet does not pull back close to the old packet.

At the chosen descriptive thresholds:

condition	count out of 120 updates
transport distance ≤ 0.25	66
primal residual ≤ 0.10	95
adjoint residual ≤ 0.25	0
both primal and adjoint residuals ≤ 0.25	0
transport distance ≥ 0.90	23

The value 0.25 is not asserted to be a universal physical threshold. It is used to make the directional imbalance visible. More strongly, even the smallest observed adjoint residual is 0.323269, so no threshold below that can pass one of these updates.

6 Scale-resolved behavior

The failure is not confined to one outlier scale.

σ	updates	median distance	max. distance	max. primal	min. adjoint
0.16	8	0.3132	0.9325	0.0301	0.3233
0.08	12	0.0440	0.4251	0.0403	0.4136
0.04	22	0.0687	1.0000	0.2552	0.4484
0.02	34	0.3972	0.9997	0.4187	0.4808
0.01	44	0.2618	1.0000	0.7219	0.4415

At $\sigma = 0.08$, primal transport is especially accurate, yet the adjoint wall remains. At finer scales both the worst primal residual and the largest packet jumps increase. Five anchors cannot establish asymptotic monotonicity, but they do not support an emerging two-sided reducing bundle.

7 Secondary diagnostics: conditioning, side, and update age

Several descriptive correlations help distinguish numerical conditioning from genuine directional geometry. They are computed on the archived finite rows and are not inferential statistics.

First, the Pearson correlation between $\log_{10} \kappa_t$ and the logarithm of the direct-formula isometry defect is 0.9840. This near-linear relation confirms that the loss in (1) is governed primarily by inverse-square-root conditioning. It also explains why the stable SVD formula repairs the isometry without changing the selected range.

Second, the transport distance correlates more strongly with the primal residual (0.6500) than with the adjoint residual (0.3358). The primal and adjoint residuals themselves have correlation 0.5183. They share some difficult updates but are far from interchangeable.

The two physical channels show different asymmetries:

side	updates	median distance	median primal	median adjoint
left	60	0.1285	0.03287	0.76172
right	60	0.3064	0.02675	0.58765

The right channel transports farther in median distance but has a smaller median adjoint defect. Therefore no single scalar "coherence score" orders the channels consistently.

Finally, the first two updates after each seed are substantially harder than later updates. Across the 20 early updates, the median transport distance is 0.9594, the median primal residual is 0.0854, and the median adjoint residual is 0.9235. Across the remaining 100 updates, the corresponding medians are 0.1455, 0.02447, and 0.7033. This suggests that a delayed start or block warm-up deserves study, but the persistent later adjoint median shows that warm-up alone does not close the two-sided gate.

Remark 7.1 (Why these correlations do not upgrade the audit)*.* The rows share scales, operators, and time histories, so they are strongly dependent. The correlations summarize mechanisms inside this archive; they are not confidence intervals, asymptotic exponents, or evidence for a population law.

8 What the negative result does and does not say

The finite data reject the following immediate inference:

$$\text{independent top-Gram resets} \Longrightarrow \text{two-sided approximately reducing history bundle.}$$

That inference was never a theorem, and RH-173 supplied an abstract exact counterexample. RH-174 shows that the concern is active in the physical frozen models rather than merely pathological.

The result does *not* reject:

1. a lagged, block, or smoothed packet schedule;
2. a Petrov–Galerkin construction with distinct primal and dual packets;
3. a finite cyclic closure in which time orientation is stored by a wrap edge rather than consecutive reset invariance;
4. a direct transfer-space Riesz construction that bypasses source memory;
5. a different topology or an all-level theorem not captured by these five scales.

It also does not invalidate the RH-151 reset certificates. Those certificates establish packet selection at each snapshot; they do not claim temporal reduction.

9 Updated status of the physical realization leaf

The refined factorization remains

$$X_{\text{phys}} = X_{\text{mem}\to\text{hist}} \wedge X_{\text{hist}\to\text{transfer}}.$$

The first factor is proved exactly and numerically stable. A particularly simple candidate for the second factor–identify consecutive reset packets as a two-sided history cocycle bundle and then map that bundle onward–is not supported by the current audit. The next theoretical step should therefore study the square infinite-history completion and determine whether it can carry a determinant. If the weighted shift makes that impossible, a finite cyclic closure becomes the natural alternative.

The audit is not interval arithmetic. No outward residual bound, physical history-to-transfer map, Riesz contour, all-level asymptotic estimate, Schatten complement, canonical determinant, Hilbert–Polya operator, zeta identity, or Riemann Hypothesis is proved.

References

[1] Liang Wang. Ky–fan reset packet atlas, 2026. Prime Dynamics Theory Program, RH-151.

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[3] Liang Wang. The exact normalized history cocycle, 2026. Prime Dynamics Theory Program, RH-173.

[4] Liang Wang. Postblock effective-rank compression, 2026. Prime Dynamics Theory Program, RH-77.

[5] Liang Wang. Source-seeded four-direction horizon refresh, 2026. Prime Dynamics Theory Program, RH-94.